

EEE 206 (January 2023)

Energy Conversion II Laboratory

Final Project Report

Section: A2 Group: **05**

3-Phase Linear Induction Motor (2-Pole & 4-Pole)

Course Instructors:

Ramit Dutta, Lecturer, EEE, BUET

Tashfiq Ahmed, Lecturer, EEE, BUET

Signature of Instructor: _____

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Signature: _____ Full Name: S.M Mahfuz Student ID: 2006060	Signature: _____ Full Name: Md. Zubaer Hossain Student ID: 2006061
Signature: _____ Full Name: Farhan Nadim Student ID: 2006062	Signature: _____ Full Name: Afnan Ali Student ID: 2006063
Signature: _____ Full Name: Anas Rohan Student ID: 2006064	Signature: _____ Full Name: Md. Shakib Hossain Student ID: 2006065

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1 Abstract :

A Linear Induction Motor (LIM) is a specific type of alternating current (AC), multiple-phase machine that provides force and movement in a linear direction. Numerous applications of LIM's can be found in industry today, one of the most interesting being high speed magnetic levitation railway systems. Application of linear induction motor includes various industrial processes like conveyor belt, vertical movement, steel power plant, induction heating and automated material handling. A LIM operates under the same principles as its AC rotary motor counterpart, typically powered by a three-phase voltage scheme with a force that is produced by a moving magnetic field. In this project a 3-Phase Linear motor of 2 & 4 Pole is built from scratch.

2 Introduction :

A Linear Induction Motor (LIM) is a special type of motor used to achieve rectilinear motion rather than rotational motion as in the case of conventional motors. A LIM operates under the same principles as its AC rotary motor counterpart, typically powered by a three-phase voltage scheme with a force that is produced by a moving magnetic field. The main difference between a typical rotary motor and a LIM is the way the magnetic field is produced. Inside of an AC rotary motor the magnetic field produced is in an infinite loop rotary motion. A LIM, however, does not operate in an infinite magnetic field loop.

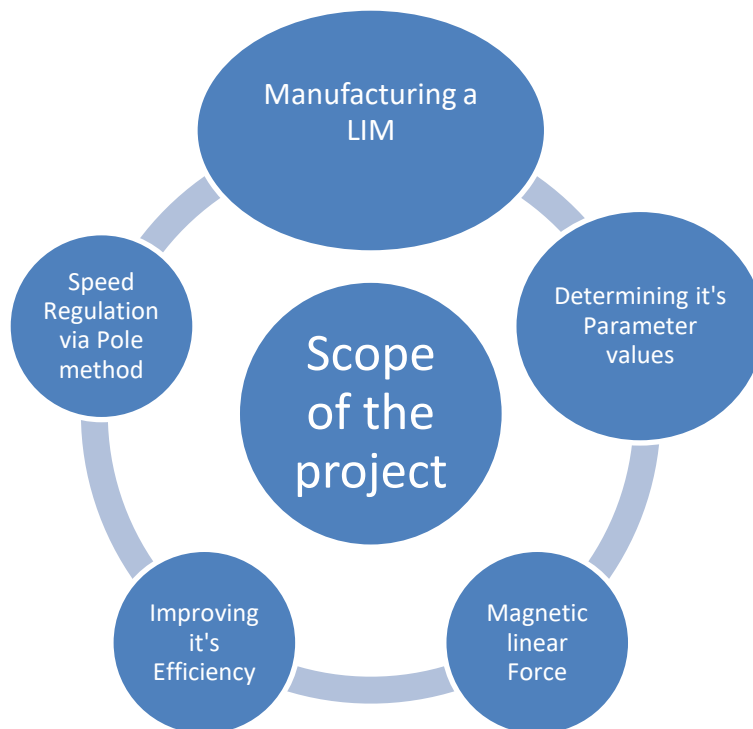
A LIM can be imagined as an AC rotary motor with the loop cut down the center and the stator and rotor spread out on a flat line, the induced magnetic field now moves linearly across the flat motor face instead of rotating. Force is achieved in a linear direction and it is understandable that this would have certain applications in industry, such as rotating large radius wheels and in industrial conveying machines. Very high speeds are obtainable because of minimal friction losses due to the magnetic levitation of the system. However, there are end-effects that reduce the motor's force, and it is often not possible to fit a gearbox to trade off force and speed. Linear induction motors are thus frequently less energy efficient than normal rotary motors for any given required force output. We can control the speed of the motor by changing the poles.

In this project, we have tried to construct a linear induction motor. The project also contains the speed control method of the motor and other parameters of the equivalent circuit. Through this project, we tried to get the working and future possibilities of a LIM.

3 Design

a. Problem Formulation (PO(b))

i. Identification of Scope



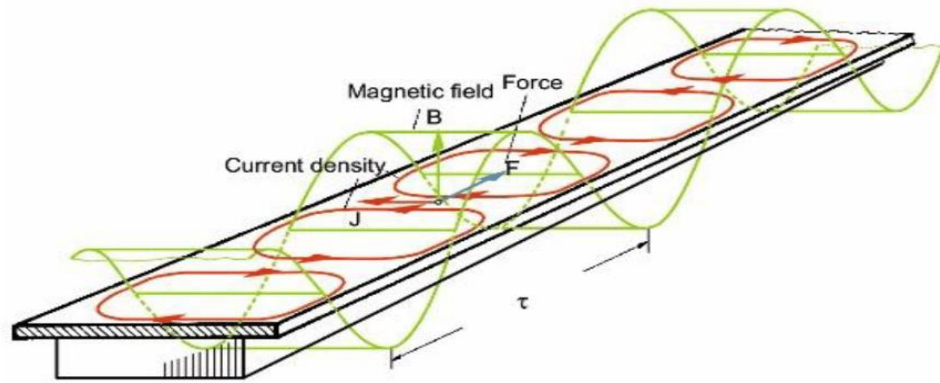
ii. Formulation of Problem

Working Principle of LIM:

Linear Induction Motor is built on the idea of linearly moving magnetic field. This magnetic field exerts a linear force on an aluminum plate due to eddy current. We can divide the working principle into 3 sections –

1. Generation of linearly moving magnetic field:

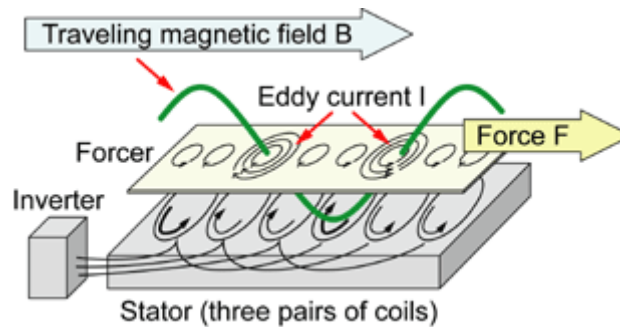
We have placed a three-phase winding connection on a iron core and connected it to a three phase supply. When the AC power is applied to the stator coils, they produce a magnetic field that alternates in polarity along the length of the stator. This alternating magnetic field interacts with the rotor, inducing eddy currents in the rotor material due to Faraday's law of electromagnetic induction.



(b)

2. Generation of Force:

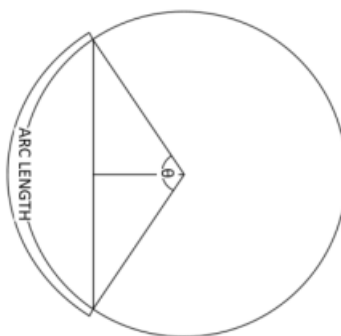
The eddy currents in the rotor create their own magnetic field. This induced magnetic field interacts with the stator's magnetic field, resulting in a repulsive or attractive force between the two fields, depending on the relative polarities. This force causes the rotor to move along the linear track.



3. Linear Motion:

As long as the stator continues to supply AC power, the magnetic fields continue to oscillate, generating a traveling magnetic wave along the stator's length. The rotor follows this magnetic wave, producing linear motion in the desired direction.

The first design steps consisted of developing a rotary to linear model.



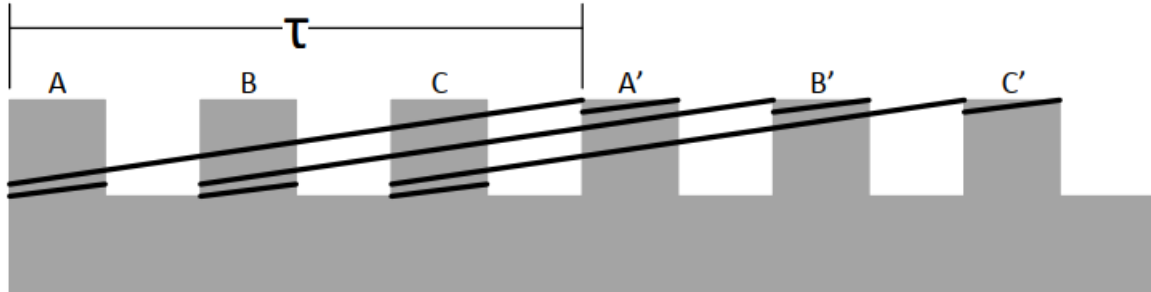
$$L = \theta r$$

Where arc length, L

For number of poles, p, and frequency f, The rotational speed of an induction motor is given by,

$$\omega = \frac{120f}{p}$$

But for a linear induction motor velocity, v , is not dictated by how many poles are associated, but rather, the pole pitch, τ , that is associated with the linear induction motor.



$$v = 2\tau f$$

$$\tau = L p$$

$$v = 2Lf p$$

1. **Synchronous speed.** It is given by $v_s = 2wf$
2. **Slip.** It is given by $s = (v_s - v) / v_s$
where v is the actual speed.
3. **Thrust or Force.** It is given by $F = P_2 / v_s$
where P_2 is the active power supplied to the rotor.
4. **Active Power Flow.** It is similar to that in a rotating motor.

(i) $P_{cr} = sP_2$ and

(ii) $P_m = (1 - s) P_2$

Speed Control by Pole Changing Method:

We used the pole changing method to change the speed of the moving aluminum plate (rotor). We know from induction motor theory that the speed of the rotating magnetic field,

$$n_s = \frac{120 * f_s}{p}$$

The same principle applies to a linear induction motor. The aluminum sheet is the rotor that follows the sliding magnetic field in this case, and for an electric cycle, the sheet slides twice the pole pitch. The distance between two poles (one north and one south) is measured in pole pitch. So, the equation of synchronous speed is

$$v = 2 \times \tau \times fs$$

where

Pole pitch τ = distance between the similar pole/2

So, if we change the number of poles in our linear induction motor from 2 pole to 4 pole, pole pitch τ gets halved so do the speed of the sliding magnetic field. Hence by changing the number of poles we can control the speed of a linear induction motor.

Change in Induced Torque of a Linear Induction Motor:

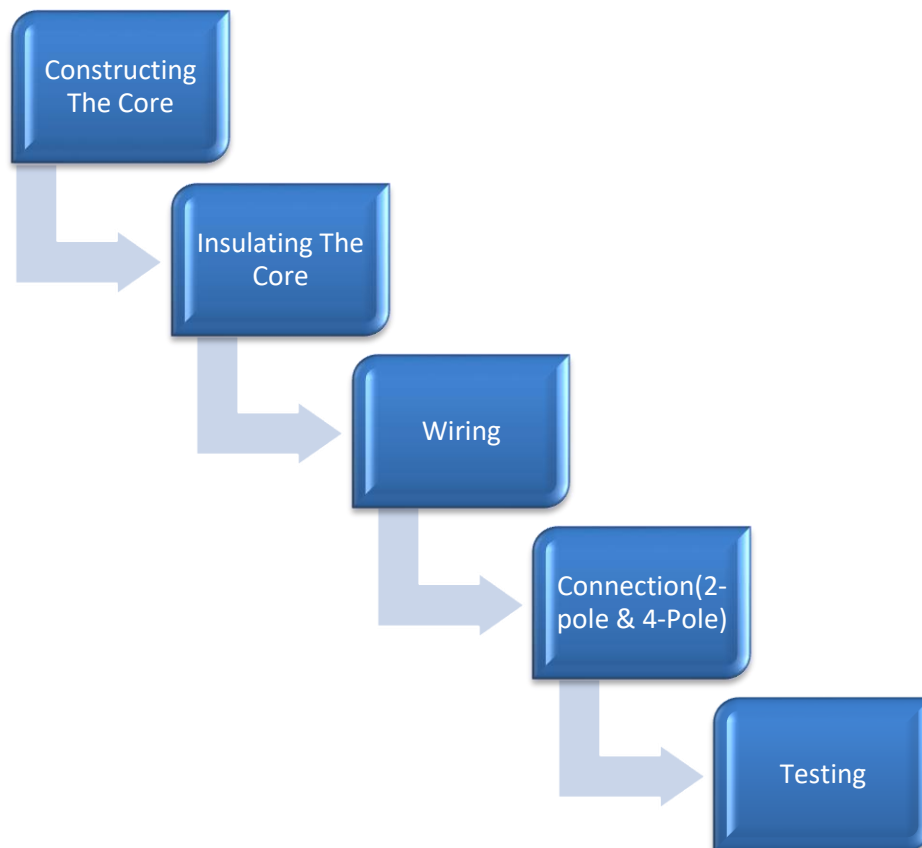
The induced torque in the rotor changes with the change in the synchronous speed, that is, the speed of the travelling magnetic field of the motor. The relation between the synchronous speed and the induced torque is given by the formula-

$$\tau = \frac{P_{AG}}{\omega_{sync}}$$

where P_{AG} is the air gap power. Air gap power is the power transferred from the machine's stator to the rotor via the air gap. The synchronous speed is obviously high in a 2-pole configuration for a constant air gap power. As a result, the induced torque is relatively low. The synchronous speed in a 4-pole configuration is half that of a 2-pole configuration. As a result, the induced torque is twice that of the two-pole configuration.

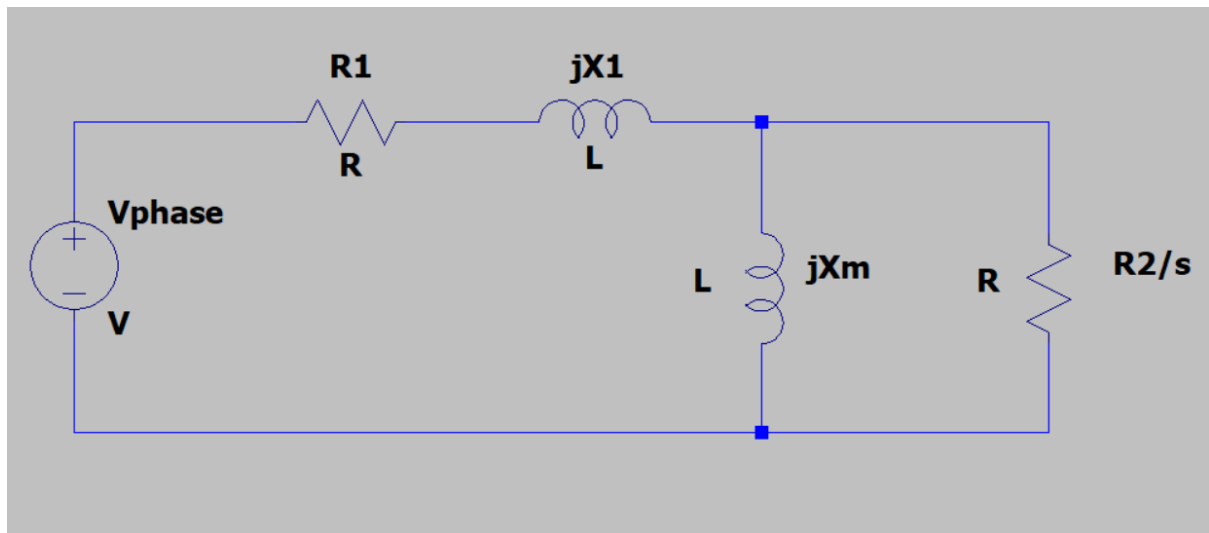
b. Design Method (PO(a))

Design Procedure:



c. Circuit Diagram

The single phase equivalent circuit for the 3-Phase Linear Induction Motor



4 Implementation

a. Description

Construction of Core :



Fig: Old Transformer

- We bought 4 old transformers and opened them up for E-Type core. We tried to have all the core of similar size as to make uniform slot and teeth. Laminated E core was used to minimize Eddy current loss.

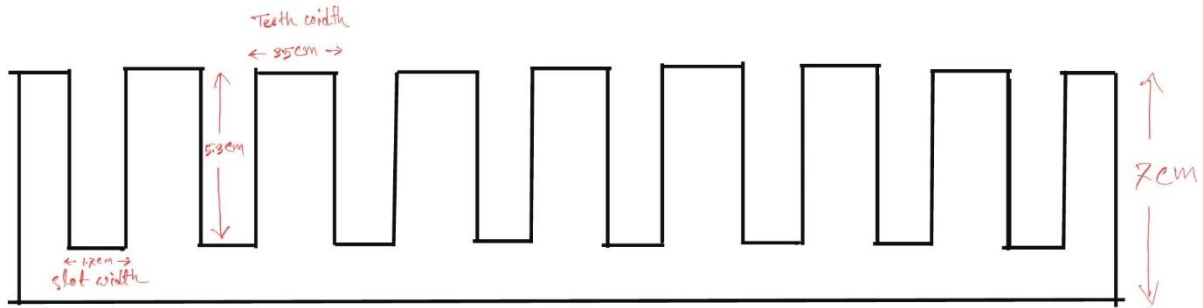


Fig: Stator Design

- We used two metal plug bars as support to attach the laminated E core and form the stator of width 4cm.
- I-type core was used to hold the plug bar and E-Type cores as a washer bolt.



Fig: Final Stator

- To prevent any unexpected current flow, we insulated the core with insulating paper. Aluminum sheet was used as the rotor

➤ **Insulating the Core:**

- Insulating paper was used to avoid connection risk.
- Tape was used to hold and insulate the shorted Copper wire connections.

Specification Of the Stator:

Spec	Measurement
Core	E-Type
Total Stator Length	45cm
Slot Length	1.7cm
Teeth Length	3.5cm
Teeth Width	4cm
Yoke Height	2cm
Total Height	7cm

➤ Wiring:

Wire Specification:

Spec	Measurement
Type	Insulated Copper Wire
Gauge	22 Gauge
Turns Per Coil	80
Rated Current Rating	9A
Wire Cross Section Area	0.325 mm ²

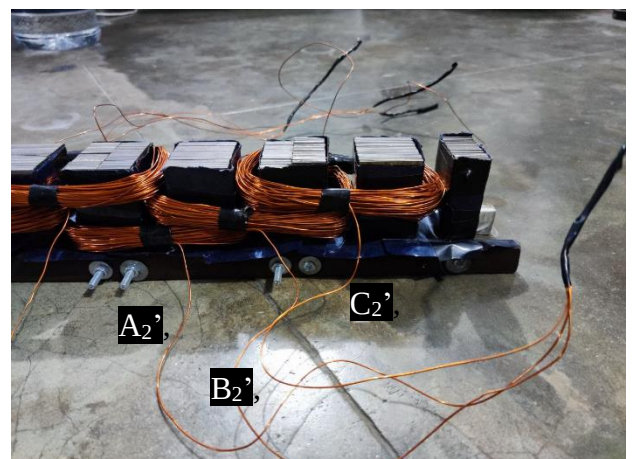
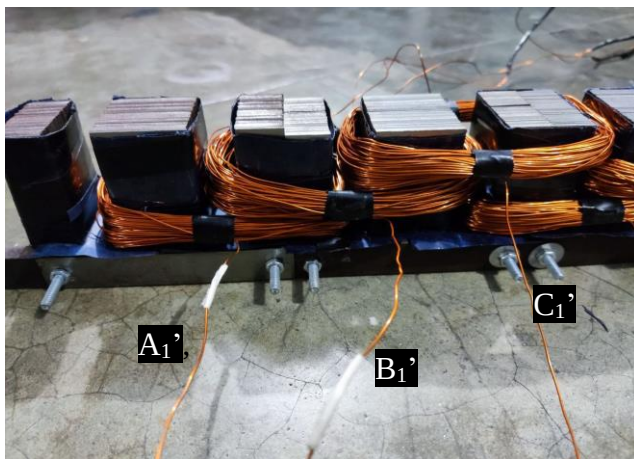
To wire the motor we use 2 set of 3 phase connection which consists of 6 coils. We made all the coil so that the wire rotates in counterclockwise manner.

First, we did the wiring of the winding on a separate wooden piece with the help of 4 nails, put at a distance of the dimension of one phase connection. Then, we placed the winding on our core.

The process of wiring was same for both 2 poles and 4 poles. Pole number differed according to our procedure of connecting the wires of two coils.



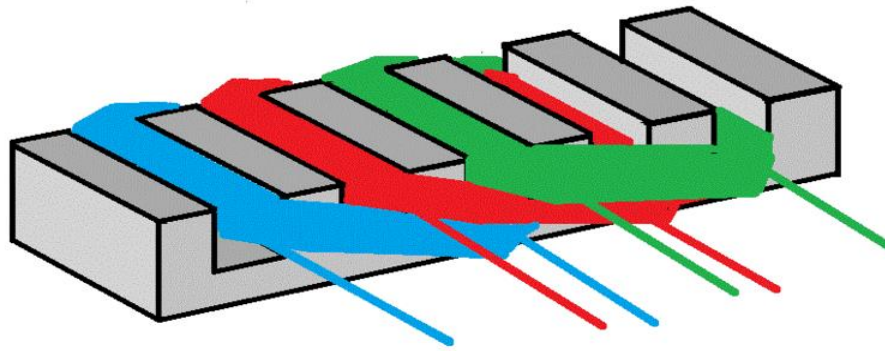
➤ Wiring and Connection:



To make a three-phase Linear Induction Motor, we had two coils per phase. 3 ends of wiring were assigned each as input or output terminals of each separate coil. The construction of 2 pole and 4 pole directly depends on the wiring of the motor.

Power supply was given to the input terminal of coil A_1' , B_1' , C_1' . Output terminal of coil-one was to connect with either terminal of coil two. The unconnected terminal i.e., output terminal of coil two was to be shorted and acted as the neutral point.

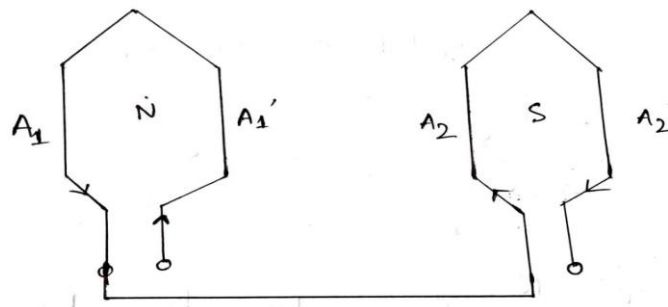
For 2-Pole connection 3-Phase Supply was given in the A_1' , B_1' , C_1' Terminals. And Neutral point was created by shorting the A_2' , B_2' , C_2' Terminals. For 4-Pole connection 3-Phase Supply was given in the A_1' , B_1' , C_1' Terminals. And Neutral point was created by shorting the A_2 , B_2 , C_2 Terminals.



Wiring Diagram For 3 coils A1,B1,C1

Connection for 2 Pole Operation:

For 2 pole operation, we need to ensure that currents in the two corresponding coils flow in the opposite direction such as if in A1 coil the current flow in the counterclockwise direction then in A2 coil the current should flow in clockwise direction. As a result, if in the A1 coil the direction the produced magnetic field is up to down then in A2 coil the direction will be from down to up. Therefore, there will be 2 opposite pole side by side. So, we will get a 2-pole configuration.

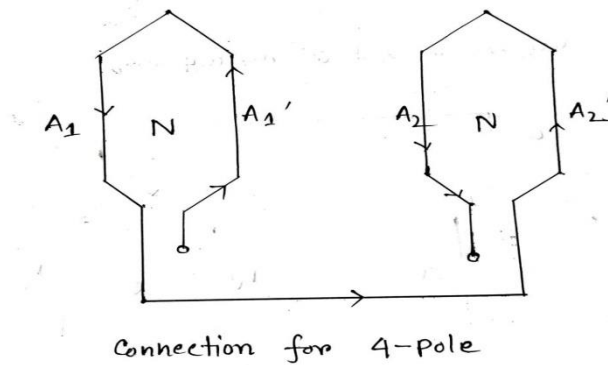


Connection for 2-pole connection

There are 6 terminals in each side of the LIM. Two of them are for same phase. If we connect the corresponding phases from the same side, then we get opposite direction of current flow in the coils. For 2-Pole connection 3-Phase Supply was given in the A_1' , B_1' , C_1' Terminals. And Neutral point was created by shorting the A_2' , B_2' , C_2' Terminals. A_1 & A_2 were shorted according to the diagram. Similarly B_1 & B_2 were shorted, and C_1 and C_2 were shorted.

Connection for 4 pole Operation:

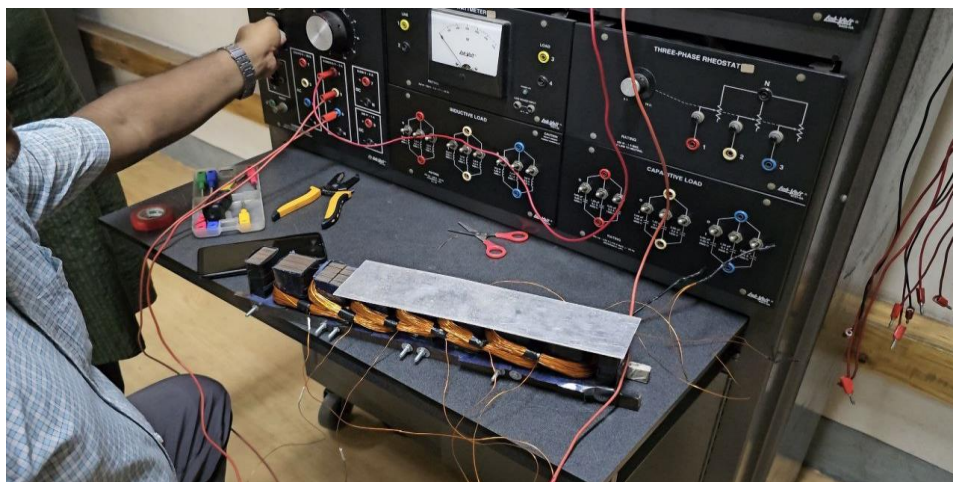
For 4 pole operation, we need to ensure that currents in the two corresponding coils flow in the same direction so that there will be 2 similar pole side by side. When there 2 strong North poles, we will get a weak South pole between them. So, we will get a 4-pole configuration. There are 6 terminals in each side of the LIM. Two of them are for same phase. If we connect the corresponding phases from the opposite side, then we get same direction of current flow in the coils.



For 4-Pole connection 3-Phase Supply was given in the A_1' , B_1' , C_1' Terminals. And Neutral point was created by shorting the A_2 , B_2 , C_2 Terminals. A_1 & A_2' were shorted according to the diagram. Similarly B_1 & B_2' were shorted, and C_1 and C_2' were shorted.

b. Experiment and Data Collection

Our Linear Induction Motor was tested in the Advanced Power Laboratory on 4/9/23. The LIM was tested for 2-Pole and 4-Pole and corresponding Voltage and Current was measured.



Line Voltage Measurement



From the AC Voltmeter ,Line Voltage = 30V

From the AC Ammeter, Line Current = 5A

For the 2-Pole Connection, according to the equation ,

$$v = 2 \times \tau \times fs$$

Here, $\tau=14\text{cm}$,

$$Vs = 2 * 0.14 * 50 = 14\text{ms}^{-1}$$

for 2-Pole Connection. But experimental speed was so much slower than that. There was high slip in the LIM.

Here, $\tau=7\text{cm}$,

$$Vs = 2 * 0.07 * 50 = 7\text{ms}^{-1}$$

for 4-Pole Connection. But experimental speed was so much slower than that. The speed of 4-Pole LIM was much lower to the comparison with 2-Pole. High slip was noticed in the experimental LIM.

c. Data Analysis

From the tests it is clear that the field current rises very quickly with the line voltage applied to the stator winding. Both for 2-pole and 4-pole, about 30V line voltage produced 5A of field current. Due to lack of facilities, the field current could not be increased further.

d. Results

In 2-pole test run 1, the rotor did not move smoothly but deflected a bit because of uneven teeth of our stator. But in the second test the applied voltage was increased therefore the field current increased and the rotor moved forward with fair amount of velocity.

In 4-pole test run 1, the rotor moved very slowly. Then in the second run the line voltage slightly increased but the rotor movement was still very slow.

We can see that the speed in 2-pole configuration is higher than the 4-pole configuration. Since torque is inversely proportionate with speed, the torque in 4-pole thus will be greater than that of 2-pole. Therefore, for higher speed we need to operate the motor in 2 pole and for higher torque with greater load we need to operate the motor in 4-pole. Also, there is a large slip in the test case due to significant air gap, which makes the P_{conv} small by causing air gap loss. Resulting in a slower speed than the speed of moving magnetic field.

e. YouTube Link

5 Design Analysis and Evaluation

Old small size transformers were collected as the main source of E type sheet. Transformers were dismantled and every E type sheets were separated. Then they were attached to build an E type core. 4 E-type cores were added linearly to make the stator. They were adjusted on plug bar using bolts and nut. In this stator there are total 9 teeth and 8 slots in between them.

Then insulation papers were to insulate the stator to insulate it from the winding. 22 gauge insulated copper cables were used for the stator winding. Winding of each coil has 80 turns. Total 6 coils were used in linear slots to complete the three-phase connection. The stator has two regions. In the first region 3 coils (can be named as A1,B1,C1) was placed and in the second region 3 more coils were placed (can be named as A2,B2, C2). Two end of cable coming out from the coil (can be named as a1-a1', b1-b1' ...) are connected differently depending on the number of pole of the motor.

a. Design Considerations (PO(c))

i. Considerations to public health and safety

- The core of the LIM is insulated with proper insulation paper to avoid any short circuits.
 - Insulated Copper wires are used for extra protection.
 - The short terminals are well taped leaving no open areas.
 - The wire has higher current rating than our system can deliver (5A).
 - Sandpaper is used to make the stator slightly smooth.
- All these are done to ensure the safety.

ii. Considerations to environment

Design consideration to environment is an approach to reduce negative environmental impact. In our project cast iron with laminated sheet are used from scrap transformer which is called recycling process. We only used new insulated wire for stator winding and aluminum sheet as rotor. So, our project fulfilled the minimum cost of production. We used all the materials which are environment friendly and don't cause pollution.

iii. Considerations to cultural and societal needs

Linear induction motor project can be implemented in those field where hand regulated linear machine are need or machines have the principle of linear motor. This project can fulfill the needs of instrumental limitation in mentioned areas. Thus, community engagement as well as job creation and economic development can be done.

b. Investigations (PO(d))

i. Data Analysis and Interpretation

Parameter Calculation:

R_1 → Per phase stator resistance. It is due to the resistive effect of the copper wire.

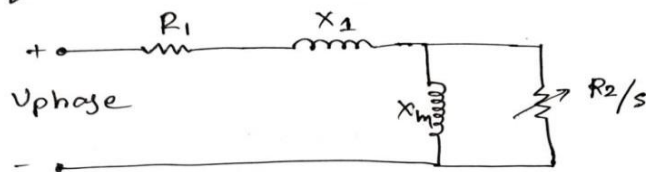
X_1 → Per phase leakage reactance. All the flux out of primary side cannot couple with the secondary. The flux loss due to air gap is known as leakage reactance. It's quite high here due to large air gap.

X_m → X_m models the effect of magnetizing current, hysteresis loss and eddy current loss. It has a very low value here to supply large magnetizing current to the core.

R_2 → Models the resistive effect of the secondary. Varies with changing aluminum sheet.

Group - 05 , ID: 2006060, 2006062, 2006064
2006061, 2006063, 2006065

Equivalent Circuit of 3-φ Linear Induction Motor



Determination of R_1 :

Per phase stator resistance, $R_1 = \frac{\rho_w L_w}{A_w t}$

$$R_1 = \frac{19.27 \times 10^{-9} \times 0.254 \times 80}{0.325 \times 10^{-6}}$$

$$R_1 = 1.2048 \Omega$$

ρ_w = Volume resistivity of copper
 $= 19.27 \times 10^{-9} \Omega m$

L = length of wire
 $= N \times \text{length of loop}$
 $= 80 \times \pi r$
 $= 80 \times \pi \times 0.254$
 $A_w t = 0.325 \times 10^{-6} m^2$

Determination of X_1 :

$$X_1 = \frac{2 \times 10^{-7} \times \left\{ \left[\lambda_s \left(1 + \frac{3}{p} \right) + \lambda_d \right] \frac{W_s t}{q_1} + \lambda_e L_e \right\} N_1^2}{P}$$

$$\lambda_s = \frac{h_s (1 + 3k_p)}{12 W_s}$$

$$\lambda_s = \frac{0.05 (1 + 3 \times 0.99)}{12 \times 0.017}$$

$$= 0.973$$

h_s = stator slot height
 $= 5 mm = 0.05 m$
 $W_s = 1.7 mm = 0.017 m$

Coil span; $\theta_p = \pi$

Pitch factor, $k_p = \sin\left(\frac{\theta_p}{2}\right) = 0.99$

$$\lambda_d = \frac{5 (g_e / W_s)}{5 + 4 \left(\frac{g_e}{W_s} \right)}$$

$$= \frac{5 \times \left(\frac{6 \times 10^{-3}}{1.7 \times 10^{-2}} \right)}{5 + 4 \left(\frac{6 \times 10^{-3}}{1.7 \times 10^{-2}} \right)} = 0.275$$

airgap
 $g_e = g_0 + d$
 $= (5 + 1) mm = 6 mm$

$\tau = 14 cm$ for 2 pole

$$L_e = \frac{\theta_p}{180} \times \tau = 0.44 \times 2.44 \times 10^{-3}$$

$\lambda_e = 0.3 (3k_p - 1)$
 $= 0.591$

$$X_1 = \frac{2 \times 10^{-7} \times 50^2 \times \left\{ \left[0.973 \left(1 + \frac{3}{2} \right) + 0.275 \right] \frac{4 \times 10^{-2}}{1.333} + 0.591 \times 2.44 \times 10^{-3} \right\} \times 80^2}{2}$$

$$= 0.0332 \Omega$$

$$\therefore X_1 = 0.0332 \Omega$$

Magnetizing reactance per phase:

$$X_m = \frac{2446 \pi f W_{seq} K_w N_1^2}{\pi^2 p g_e}$$

$$= \frac{2446 \pi \times 10^{-7} \times 50 \times 4.6 \times 10^{-2} \times 0.956 \times 80^2 \times 0.14}{\pi^2 \times 2 \times 6 \times 10^{-3}}$$

$$= 0.501 \Omega$$

$$\therefore X_m = 0.501 \Omega$$

$$k_d = \frac{\sin \alpha_1 (\alpha/2)}{a_1 \sin (\alpha/2)}$$

$$k_w = k_p \times k_d = 0.956$$

$$W_{seq} = a + \frac{b}{10} = 4.6 \text{ cm}$$

$$g_e = 6 \text{ mm}$$

Calculated Value of the circuit parameters:

Parameter	Values
R1	1.2048ohm
X1	0.033ohm
Xm	0.501ohm
R2	Load Dependent

c. Limitations of Tools

We have encountered a few constraints in certain fields while completing this project. These limitations include-

- **Manually Completed Wiring and Core Layout:** The core layout and the wiring were done manually instead of automation due to the project not being a large-scale one or an industrial one.
- **Limited number of options for core sizes:** The available core size options were limited in number. We could manage to work only with E-Type and I-Type cores.
- **Uneven cores:** The surfaces of the cores were coarse and uneven. They needed to be made smooth manually using sandpaper. Our Rotor also wasn't having a smooth path to be in motion.
- **Lack of absolute elimination of air gap:** Due to all the surfaces not being smooth, the air gap between the stator and the rotor (the metal sheet) could not be absolutely

eliminated.

- **Large magnetizing current flow:** The magnetizing current flow was quite large in magnitude.
- **Weight of rotor:** The weight of rotor is higher as a limitation of instrument. It is difficult for motor to drive the aluminum sheet with certain speed.

d. Impact Assessment

i. Assessment of Societal and Cultural Issues

The linear induction motor can grow a lot of societal issues if it is used in different fields of machineries. The societal and cultural issues can be-

1. Cultural device improvement: We have some cultural devices that are used to make our identity an unique one as a nation. Like 'hand-made' *jamdani sharee*, 'hand-controlled' *kheya* boats, man-made earthen pots, sculpture etc. are some examples of them. With the concept of linear induction motor, we can make the machines using in these works, more easier to work with (even without the artists behind them) but this artificial transform will make our cultural pride deemed.

Counter logic: Culture is not an immortal thing. With flow of time culture becomes replaced, reformed and re-innovated. May linear induction motor can have a negative change there, but if there were no linear induction motors, this negative changes will come also in other forms with the flow of time.

2. Profession loss: In case of making use of linear induction motors in different devices will make it easier to work with but we may have to lose the professionals who work behind all the common devices. We may have to replace them with engineers and technicians.

Counter logic: New technology all time degrades the present existence of professional systems but it also opens doors for those who can cope with the newer situations and improvements. So the professionals will be changed, profession will be changed but the working rate or worker number will not die out.

3. Prohibition against cultural or societal independence: In recent days, we have encountered some problems regarding the metro-rail caused by kite flying, which is nothing less than a symbol of a citizen's cultural freedom but due to interference in the power system of metro-rail we had to make cross amendments in corresponding sections of rules. The same is going to happen when we will have the intention to use linear induction motors in different mechanisms. We have to bring new laws that will restrict some of the freedom experienced by a common citizen in the present condition.

Counter logic: Laws and technologies are for the betterment of people. Maybe linear induction motors will introduce some newer restrictions but the impact of using linear induction motors is so much helpful that the impact of these restrictions are negligible with respect to the impact of the introduction of linear induction motor technology.

ii. Assessment of Health and Safety Issues

Some safety and health issues are there regarding linear induction motors like-

1. Electrical safety issues: As high voltage will be used in linear induction motors, a lot of chances are there regarding electrical accidents. So safety issues are a great source of concern.

Counter logic: The designing of a machine is all time done giving the highest priority to electrical safety. So as we designed our device to minimize electrical harm, may it can be used safely. To provide more safety, we can add additional devices in parallel with it like fuses and other control devices.

2. Rotor freezing time issue: Generally when we use a technology to enhance vehicles motion, we always try to make the speed of the vehicle controllable very quickly to prevent accidents. But if we use linear induction motors in vehicles, we may face issues to stop the rotor in a small reaction time due to high inertia of the rotor.

Counter logic: We can use an additional winding which can provide necessary field and force to stop the movement of a rotor very quickly in a less reaction time.

3. Core production health hazard: Generally core production techniques make some health issues to the workers.

Counter logic: We can try to decrease the effect by taking proper measures in corresponding arena.

iii. Assessment of Legal Issues

Some legal issues that can be encountered while using linear induction motors like-

1. Linear induction motor placement: As linear induction motors will be heavier than a general average-weighted machine, we need to design such a civil engineered frame that will not face any harm due to at least richter 7.0 earthquake. But in our country some buidings are registered as safe for operation with heavy machine, but they do not show proper result regarding this.

Counter logic: Before implanting any linear induction motor an civil engineered inspection can be made mandatory to prevent the problems regarding it.

2. Criminal issues: As we can give the rotor higher speed with a linear induction motor by improving core , turn number ,rotor and wire,it can be used for any kind of automated killing or harming missions.

Counter logic: Necessary laws can prevent people from implementing this kind of misdeeds.

e. Sustainability and Environmental Impact Evaluation (PO(g))

Sustainability evaluation: The linear induction motors are very much sustainable if we take

the rated measurements in concern while using the devices. Again, as there are not a lot of machineries in its construction, the system is easier to control and easier to keep it in its safe and life-enhancing working area.

Environmental Impact Evaluation: The linear induction motor will have a very little environmental negative impact in fields like-

1. Core production system.
2. Electricity production system

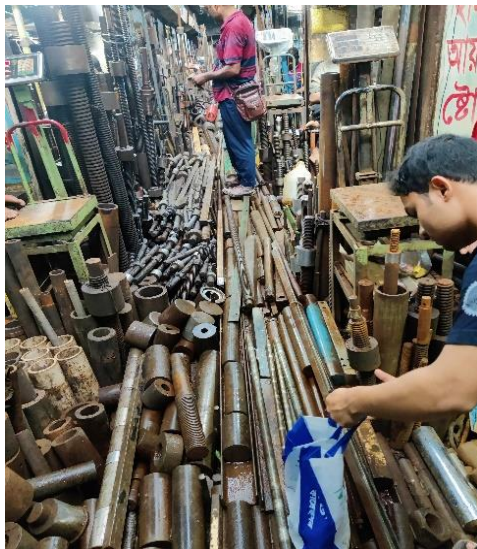
On the other hand ,directly linear induction motors have no environmental bad impact. So it can be safely used in open environment.

f. Ethical Issued (PO(h))

No kinds of unfair means was adopted during the whole project. We went to the shop (Nawabpur & Dholaikhal) and bought all the required materials. Then Built the LIM from scratch. We actively made the stator and insulated it with insulation paper. All the wiring also done by hand in a 4-nail structure. As we required six coils, every member of the group wired one coil.

6 Reflection on Individual and Team work (PO(i))

a. Individual Contribution of Each Member



Buying accordingly



Sorting



Inserting



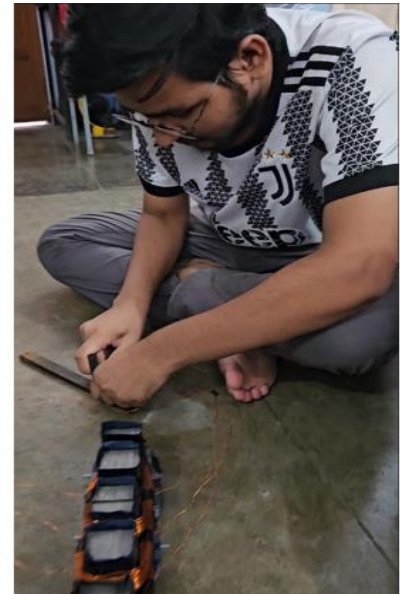
Measuring accordingly



Making The Stator



Insulating The Core



b. Mode of TeamWork

The modes of teamwork we used are:

- i) **Collaborative Teamwork:** We used this mode the most. Most of the work related to our project was conducted through collaborative teamwork. Often, we used our hall room to get together and work on the project. During this mode, we shared different opinions on how we could easily achieve something, we discussed important topics related to the project and shared our responsibilities.
- ii) **Divided Responsibilities:** In this mode, team members were assigned specific tasks. Sorting, measuring, inserting, turning the coils, sharpening connection lead etc. were assigned to each member of the group. This way project has been done before due date.
- iii) **Leader-Driven Teamwork:** To make decisions fast, get direction, delegate tasks, we used one team leader. As some are attached and residents are from different Hall, one team leader was necessary to guide all of us. And used the Hall room of our team leader. Several necessary decisions were taken quickly by our team leader.
- iv) **Virtual Collaboration:** Although we used to go to the hall room for project, we used virtual platform to see the project advancement whenever there was a necessity. It gave a linkage between us through different hall and resident.

- v) **Peer Review and Feedback:** We gave a review of every step of the project and collected feedback on how to make it more efficient and less costly. Surely, it enhanced the quality of the project. It added the effectiveness of coordination.
- vi) **Hybrid Approaches:** Sometimes collaborative work was necessary; some other times leader-driven teamwork was the only option. Overall, we used different modes of teamwork mixing them up.

c. Diversity Statement of Team

Diversity Statement of Group 5

In Group 5, we firmly believe that diversity is our strength. Our team is composed of individuals with unique backgrounds, perspective, and talents. So, diversity is the tool to excel in our project.

We are committed to promoting an inclusive and respectful environment where every team member feels valued, heard, and empowered.

In our collaborative efforts, we vow to,

1. **Respect Difference:** We respect and honor the unique qualities that each team member brings to table, understanding that diversity includes not only visible characteristics but also different perspectives and approaches to problem-solving.
2. **Open Communication:** We encourage open and honest communication among team members. We believe that sharing different views leads to better decision-making and more robust outcomes.
3. **Learn from Each Other:** Diversity provides a valuable learning opportunity. We are committed to learning from each other and growing both personally and professionally through collaboration.
4. **Leverage Our Strengths:** We understand that our diverse skills and experiences are assets that can be utilized to solve complex challenges effectively. We will leverage these strengths to excel in our project.
5. **Encourage Inclusivity:** We actively promote an inclusive atmosphere where all team members have an equal opportunity to contribute, regardless of their background or experience.

We are excited to embark on this journey together, confident that our diversity will be a driving force behind our success.

Sincerely,

Group 5

d. Log Book of Project Implementation

Name	Start Date	End Date
Initial Design	Jul 24, 2023	Aug 03, 2023
Buying Parts	Aug 04, 2023	Aug 04, 2023
Second Time Buying	Aug 16, 2023	Aug 16, 2023
Building The Stator	Aug 18, 2023	Aug 23, 2023
Insulating	Aug 21, 2023	Aug 23, 2023
Wiring & Connection	Aug 24, 2023	Aug 28, 2023
Testing	Sep 04, 2023	Sep 04, 2023

7 Communication (PO(j))

a. Executive Summary

The main purpose of this project is to design and build a linear induction motor (LIM) that can generate motion and force in a linear direction. The motivation for this project originates from the lack of machines that produce linear force and motion in industry. There are many applications in industry where linear motion is necessary but unobtainable. An instance where linear motion is required is on specific assembly lines that need to move and control equipment in a flat linear fashion. Providing a solution via linear motors in these respects could increase industrial production, create an ease of control, and generates the possibility to make work environments safer and more efficient.

The LIM's stator is created using copper wires for coils and E-Type stator lamination segments for the stator. The stator is powered by a three-phase voltage source and the rotor is an aluminium sheet to show how a LIM produces linear motion. For controlling the speed of the motor, Pole controlling method is used. Here we used 2-Pole and 4-Pole for different speed. Poles are changed by changing the connections. The performance of the motor is tested in the Advanced Power Laboratory. The Advanced Power Laboratory offers access to a three phase voltage scheme.

As our project is in small scale, we couldn't ensure a good efficiency for the motor. The stators are made from old transformers E-Type stator as a result they are uneven in size and provides different reluctance throughout. Also the wiring is done fully by hand. If the perfect materials are ensured and modern machinery are used, efficiency can be increased.

b. User Manual

The motor has total 6 cables coming out from the stator coil. The connections depend on number of poles.

2-pole operation: For 2-pole operation, short A1 with A2, B1 with B2 and C1 with C2 cable. Then short A2', B2' and C2' to make the neutral. To add the power supply, connect three-

phase connection with A1', B1' and C1'. To measure line voltage, connect a voltmeter of 0-250V range parallel with A2' and B2'. To measure field current, connect an ammeter with minimum range of 0-5A in series with the line.

4-pole operation: For 2-pole operation, short A1 with A2', B1 with B2' and C1 with C2' cable. Then short A2, B2 and C2 to make the neutral. To add the power supply, connect three-phase connection with A1', B1' and C1'. To measure line voltage, connect a voltmeter of 0-250V range parallel with A2' and B2'. To measure field current, connect an ammeter with minimum range of 0-5A in series with the line.

Finally place the rotor (aluminium sheet) on top of the stator. Turn on the power supply and slowly increase the voltage until 5A of current flows through the field.

Caution:

Be careful about the connection and the insulation of the wire and stator. Damaging the insulation may cause catastrophic failure in the motor. The line voltage should be given according to the limit of field current otherwise the ammeter or the stator winding can be burnt off. Use shock proof gloves and sandals made of non-conducting material to avoid any kind of accident. The power should be turned off after completing the tests.

8 Project Management and Cost Analysis

a. Bill of Materials

Materials	Price (BDT)
E-Type Core	1900
22 Gauge Copper Wire (1kg)	1230
Plug Bar	170
Insulation Paper	30
Nut, Screw	70
Tape	50
Aluminium Sheet	100

b. Calculation of Per Unit Cost of Prototype

Materials	Price (BDT)
E-Type Core	1900
Wire 22 Gauge (1kg)	1230
Plug Bar*2	350
Insulation Paper	30
Nut, Screw	70
Tape	50
Aluminium Sheet	100
Total	3730

c. Timeline of Project Implementation

ID	Name	Jul, 2023			Aug, 2023				Sep, 2023	
		17 Jul	23 Jul	30 Jul	06 Aug	13 Aug	20 Aug	27 Aug	03 Sep	1...
1	Initial Design									
2	Buying Parts									
3	Second Time Buying									
4	Building The Stator									
5	Insulating									
6	Wiring & Connection									
7	Testing									

9 Future Work (PO(I))

This project acts as the foundation of a lot of potential work that can be done. Some of them are-

- Without controlling the speed manually by pole changing, Solid state drives (VFD) can be made for speed control with the help of linear induction motors by controlling their frequencies.
- For better performance, the wiring and core layout should be automated rather than done by hand, especially if we can create the cores ourselves or choose a specific design for a given application.
- Developing switching processes using linear induction motor.

10 References

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